

Program : Diploma in Architecture	
Course Code : 3188	Course Title: Masonry and Services Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

This course is designed to:

- Identify the quality of construction materials, and understand their relevant characteristics
- Identify the suitability of various materials for different construction process.
- Set out a building at field from building plan
- Supervise construction of brick wall, electrical and plumbing works in buildings.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Engineering Mathematics	1
Knowledge about basic Physics		Applied Physics I	1
Basic knowledge about building materials		Construction materials and engineering	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Conduct tests to find out the strength of different building materials related to construction activities and identify the design mix of concrete.	12	Applying
CO2	Set out the building at site from given building plan	5	Applying
CO3	Supervise construction of brick wall.	12	Applying
CO4	Supervise electrical and plumbing works in a building.	10	Applying
	Series Test	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3				2		3
CO3	3						2
CO4	3						2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Conduct tests to find out the strength of different building materials related to construction activities and identify the design mix of concrete. Interpret the strength of different building materials related to construction activities. Identify the design mix of concrete related to given construction activities		
M1.01	Determine the fineness of cement	2	Applying
M1.02	Determine the standard consistency	1	
M1.03	Determine the initial and final setting time of cement	2	Applying
M1.04	Determine the workability of concrete (slump test and compaction factor test)	2	Applying
M1.05	Conduct the compression test on brick and find the stress at first crack and ultimate stress	2	Applying
M1.06	Conduct tension test on mild steel rod and find yield point stress, breaking stress, ultimate stress, percentage of elongation and percentage of reduction in constructional area of MS bar	2	Applying
M1.07	Determine fineness modulus on fine and coarse aggregate	1	Applying
CO2	Set out the building at site from given building plan		
M2.01	Conduct setting out of a building on site	3	Applying

M2.02	Identify various layers and types of soil in foundation pit by visiting at least 3 construction sites in different locations of city and prepare report consisting photographs and samples.	2	Applying
	Series Test I	3	
CO3	Supervise construction of brick wall.		
M3.01	Construct one brick thick wall in English bond	3	Applying
M3.02	Construct one and half brick thick wall in English bond	3	Applying
M3.03	Construct one brick thick wall in Flemish bond	3	Applying
M3.04	Construct one and half brick thick wall in Flemish bond	3	Applying
CO4	Supervise electrical and plumbing works in a building.		
M4.01	Do electrical wiring connections to lamps in series and parallel circuits and connect a two-way switch in a circuit to operate lamp.	4	Applying
M4.02	Install plumbing work for single tap and bath room shower	3	Applying
M4.03	Install plumbing work for water closet	3	Applying
	Series Test-II	3	

Text / Reference:

T/R	Book Title/Author
T1	Gambhir, M.L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd., Delhi.
R2	Shetty, M.S., Concrete Technology, S. Chand and Co. Pvt. Ltd., Ram Nagar, Delhi.
R3	Santhakumar, A. R., Concrete Technology, Oxford University Press, New Delhi.
R4	Neville, A. M. and Brooks, J.J., Concrete Technology, Pearson Education Pvt. Ltd.
R5	Sood, H., Kulkarni P. D., Mittal L. N., Laboratory Manual in Concrete Technology, CBS Publishers, New Delhi.

Online Resources:

Sl.No	Website Link
1	http://www.sginstitute.in/Downloads/Civil_Downloads/Manual_ConcreteTech.pdf www.sginstitute.in › Downloads › Civil_Downloads › Manual_ConcreteTech
2	https://lecturenotes.in/notes/2676-notes-for-concrete-technology-ct-by-madhusmita-mishra
3	https://theconstructor.org/practical-guide/material-testing/tension-test-steel-rod/2482/
4	https://www.youtube.com/watch?v=7dN3lku0Bns
5	https://www.google.com/search?xsrf=ACYBGNQag6lVf2llor8A41Wq7nmreqQzEw%3A1576216309137&lei=9SbzXYT-B76d4-EpoMe1yAk&q=concrete%20technology%20notes%20nptel&ved=2ahUKEwil6J6q97HmAhhV7xDgGHZ1XDOWQsKwBKAB6BAgBEAE&biw=1366&bih=657#